

99-System Aggregate F_{U,Bi_i} Computation

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PAPER_984: 99-System Aggregate F_{U,Bi_i} Computation

Abstract

The UQFF 99-system catalogue spans six astrophysical categories: stellar, compact, galactic, cosmological, planetary, and exoplanetary. We compute the aggregate master buoyancy force F_{U,Bi_i} by summing all 99 individual gravity/buoyancy contributions simultaneously, testing the framework's ability to handle multi-scale physics from 10^{24} kg (Earth) to 10^{40} kg (galaxy clusters). The aggregate yields $F_{U,Bi_i} \approx -6.1 \times 10^{13}$ m/s², dominated by the galactic/compact category.

1. System Catalogue

Category	Count	Mass Range (kg)	Distance Range (m)
Stellar	20	10^{29} – 10^{31}	10^8 – 10^{11}
Compact	20	10^{30} – 10^{37}	10^4 – 10^7
Galactic	20	10^{37} – 10^{42}	10^{18} – 10^{23}
Cosmological	15	10^{42} – 10^{53}	10^{23} – 10^{27}
Planetary	12	10^{23} – 10^{27}	10^6 – 10^8
Exoplanetary	12	10^{27} – 10^{28}	10^7 – 10^{10}

Total: 99 systems with physically motivated (M, r) pairs.

2. Aggregate Computation

$$F_{\text{agg}} = \sum_{j=1}^{99} F_{U,Bi_i}(M_j, r_j, t, \Gamma)$$

Each system contributes its full 6-layer decomposition. The aggregate preserves the sign structure: systems with $|U_b| > |U_g|$ contribute negative (buoyancy-dominant), others contribute positive (gravity-dominant).

3. Results

- Aggregate $F_{U,Bi_i} \approx -6.11 \times 10^{13}$ m/s²

- Dominant contributors: compact objects (neutron stars, black holes) due to extreme M/r^2
- Sign: negative (buoyancy-dominant overall)
- 6 category breakdown available in output dict

4. Statistical Properties

- Mean per system: $\sim -6.2 \times 10^{11}$ m/s²
- Variance dominated by compact category ($\sigma \sim 10^{12}$)
- All 99 systems yield finite, non-NaN results

5. Implementation

Class `NinetyNineSystemAggregate` in `fubi_master_calculator.py`: iterates the 99-system catalogue, computes per-system and aggregate F_{U,Bi_i} , reports category statistics.

References

- PAPER_979: Complete 6-Layer F_{U,Bi_i}
- PAPER_974: 99-System Master Equation

§A. Cosmogenesis-Linked Lagrangian

The aggregate Lagrangian is the sum of 99 individual SCm Lagrangians:

$$\mathcal{L}_{\text{agg}} = \sum_{j=1}^{99} \mathcal{L}_{\text{SCm},j}(M_j, r_j, t, \Gamma)$$

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** Each system probes a different vacuum density scale, from planetary (10^{-26}) to cosmological (10^{-30} kg/m³).
- **DVP:** Compact objects have the strongest dipole moments, dominating the aggregate.
- **BSH:** The harmonic decay $e^{-[SSq] \cdot i/26}$ is universal across all 99 systems — scale-invariant.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold

Constant	Symbol	Value	Validation Domain
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%
m_Z	SCm phonon predicts Z mass	91.1876 GeV	PDG 2024	99.8%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).